

Prüfbericht-Nr.: Test report no.:	CN2532PX 001	Auftrags-Nr.: Order no.:	178220936	Seite 1 von 28 Page 1 of 28
Kunden-Referenz-Nr.: Client reference no.:	2672284	Auftragsdatum: Order date:	2025-05-23	
Auftraggeber: Client:	Fuzhou Tianyu Electric Co., Ltd No.28 Yaoxi Road, Nanyu Town, Minhou County, Fuzhou City, 350109 Fujian P.R. China			
Prüfgegenstand: Test item:	A Three-phase Oil-immersed Power Transformer			
Bezeichnung / Typ-Nr.: Identification / Type no.:	S-M-630/22-Tier2			
Auftrags-Inhalt: Order content:	Power Transformer Efficiency Measurement or Calculation			
Prüfgrundlage: Test specification:	EN 50708-2-1:2020, Directive 2009/125/EC, EN 60076-1:2011			
Wareneingangsdatum: Date of sample receipt:	2025-05-26			
Prüfmuster-Nr.: Test sample no.:	BY04101			
Prüfzeitraum: Testing period:	2025-05-27			
Ort der Prüfung: Place of testing:	Shandong Institute for Product Quality Inspection			
Prüflaboratorium: Testing laboratory:	Inspected by TÜV Rheinland (Shanghai) CO. Ltd.			
Prüfergebnis*: Test result*:	Pass			
geprüft von: Tested by: Ming Sun	<i>Zehua Song</i> Trainee: Zehua Song <i>Ming Sun</i>	genehmigt von: authorized by: Jingsong Xu	<i>JingSong Xu</i>	
Datum: Date:	2025-07-10	Ausstellungsdatum: Issue date:	2025-07-10	
Stellung / Position:	Project Engineer	Stellung / Position:	Authorizer	
Sonstiges / Other:	This oil-immersed power transformer S-M-630/22-Tier2 had already passed the type tests, which conform to IEC 60076-1, IEC 60076-2, IEC 60076-3, IEC 60076-5, IEC 60076-10, details to see the type test report CN25ZJN4 001 issued by TÜV Rheinland. Transformer with rated power $S_r \leq 3150$ kVA, the maximum level of losses for no-load losses and for load losses is class TIER2, meet the efficiency requirements of the standard EN 50708-2-1:2020 and comply with the Energy Efficiency Directive 2009/125/EC.			
Zustand des Prüfgegenstandes bei Anlieferung: Condition of the test item at delivery:	Prüfmuster vollständig und unbeschädigt Test item complete and undamaged			
* Legende:	P(ass) = entspricht o.g. Prüfgrundlage(n)	F(ail) = entspricht nicht o.g. Prüfgrundlage(n)	N/A = nicht anwendbar	N/T = nicht getestet
* Legend:	P(ass) = passed a.m. test specification(s)	F(ail) = failed a.m. test specification(s)	N/A = not applicable	N/T = not tested
Dieser Prüfbericht bezieht sich nur auf das o.g. Prüfmuster und darf ohne Genehmigung der Prüfstelle nicht auszugsweise vervielfältigt werden. Dieser Bericht berechtigt nicht zur Verwendung eines Prüfzeichens. <i>This test report only relates to the above mentioned test sample. Without permission of the test center this test report is not permitted to be duplicated in extracts. This test report does not entitle to carry any test mark.</i>				

Prüfbericht-Nr.: CN2532PX 001
Test report no.:

Seite 2 von 28
Page 2 of 28

Anmerkungen
Remarks

- | | |
|---|--|
| 1 | <p>Alle eingesetzten Prüfmittel waren zum angegebenen Prüfzeitraum gemäß eines festgelegten Kalibrierungsprogramms unseres Prüfhauses kalibriert. Sie entsprechen den in den Prüfprogrammen hinterlegten Anforderungen. Die Rückverfolgbarkeit der eingesetzten Prüfmittel ist durch die Einhaltung der Regelungen unseres Managementsystems gegeben.</p> <p>Detaillierte Informationen bezüglich Prüfkonditionen, Prüfequipment und Messunsicherheiten sind im Prüflabor vorhanden und können auf Wunsch bereitgestellt werden.</p> <p><i>The equipment used during the specified testing period was calibrated according to our test laboratory calibration program. The equipment fulfils the requirements included in the relevant standards. The traceability of the test equipment used is ensured by compliance with the regulations of our management system. Detailed information regarding test conditions, equipment and measurement uncertainty is available in the test laboratory and could be provided on request.</i></p> |
| 2 | <p>Wie vertraglich vereinbart, wurde dieses Dokument nur digital unterzeichnet. Der TÜV Rheinland hat nicht überprüft, welche rechtlichen oder sonstigen diesbezüglichen Anforderungen für dieses Dokument gelten. Diese Überprüfung liegt in der Verantwortung des Benutzers dieses Dokuments. Auf Verlangen des Kunden kann der TÜV Rheinland die Gültigkeit der digitalen Signatur durch ein gesondertes Dokument bestätigen. Diese Anfrage ist an unseren Vertrieb zu richten. Eine Umweltgebühr für einen solchen zusätzlichen Service wird erhoben. Informationen zur Verifizierung der Authentizität unserer Dokumente erhalten Sie auf folgender Webseite: go.tuv.com/digital-signature</p> <p><i>As contractually agreed, this document has been signed digitally only. TUV Rheinland has not verified and unable to verify which legal or other pertaining requirements are applicable for this document. Such verification is within the responsibility of the user of this document. Upon request by its client, TUV Rheinland can confirm the validity of the digital signature by a separate document. Such request shall be addressed to our Sales department. An environmental fee for such additional service will be charged. For information on verifying the authenticity of our documents, please visit the following website: go.tuv.com/digital-signature</i></p> |
| 3 | <p>Prüfklausel mit der Note * wurden an qualifizierte Unterauftragnehmer vergeben und sind unter der jeweiligen Prüfklausel des Berichts beschrieben.</p> <p>Abweichungen von Prüfspezifikation(en) oder Kundenanforderungen sind in der jeweiligen Prüfklausel im Bericht aufgeführt.</p> <p><i>Test clauses with remark of * are subcontracted to qualified subcontractors and described under the respective test clause in the report.</i></p> <p><i>Deviations of testing specification(s) or customer requirements are listed in specific test clause in the report.</i></p> |
| 4 | <p>Die Entscheidungsregel für Konformitätserklärungen basierend auf numerischen Messergebnissen in diesem Prüfbericht basiert auf der "Null-Grenzwert-Regel" und der "Einfachen Akzeptanz" gemäß ILAC G8:2019 und IEC Guide 115:2021, es sei denn, in der auf Seite 1 dieses Berichts genannten angewandten Norm ist etwas anderes festgelegt oder vom Kunden gewünscht. Dies bedeutet, dass die Messunsicherheit nicht berücksichtigt wird und daher auch nicht im Prüfbericht angegeben wird. Zu weiteren Informationen bezüglich des Risikos durch diese Entscheidungsregel siehe ILAC G8:2019.</p> <p><i>The decision rule for statements of conformity, based on numerical measurement results, in this test report is based on the "Zero Guard Band Rule" and "Simple Acceptance" in accordance with ILAC G8:2019 and IEC Guide 115:2021, unless otherwise specified in the applied standard mentioned on Page 1 of this report or requested by the customer. This means that measurement uncertainty is not taken in account and hence also not declared in the test report. For additional information to the resulting risk based of this decision rule please refer to ILAC G8:2019.</i></p> |

Test report No.	CN2532PX 001
Order No.	178220936
Order date	2025-05-23
Client	Fuzhou Tianyu Electric Co., Ltd
Address	No.28 Yaoxi Road, Nanyu Town, Minhou County, Fuzhou City, 350109 Fujian P.R. China
Manufacturer	Fuzhou Tianyu Electric Co., Ltd
Manufacturer address	No.28 Yaoxi Road, Nanyu Town, Minhou County, Fuzhou City, 350109 Fujian P.R. China
Trademark	
Manufacture year	2025-05
Tested samples/ Tested items	A Three-phase Oil-immersed Power Transformer
Type No. / Identification	S-M-630/22-Tier2
Standards / Specifications	EN 50708-2-1:2020
Test result	Pass

Remarks:

This oil-immersed power transformer S-M-630/22-Tier2 had already passed the type tests, which conform to IEC 60076-1, IEC 60076-2, IEC 60076-3, IEC 60076-5, IEC 60076-10, details to see the type test report CN25ZJN4 001 issued by TÜV Rheinland.

Transformer with rated power $S_r \leq 3150$ kVA, the maximum level of losses for no-load losses and for load losses is class TIER2, meet the efficiency requirements of the standard EN 50708-2-1:2020 and comply with the Energy Efficiency Directive 2009/125/EC.

The assessment of energy efficiency is shown on pages 8 to 16.

The energy efficiency measurement and calculation are shown on pages 17 to 25.

Test report

Report Number: CN2532PX 001

Participants

Witness information

Person attending the test

Mr. Zehua Song Shandong Institute for Product Quality Inspection

Witnessed and/or inspected by

Mr. Jingsong Xu TÜV Rheinland (ShangHai) Co., Ltd.

Mr. Ming Sun TÜV Rheinland (ShangHai) Co., Ltd.

Mr. Zehua Song TÜV Rheinland/CCIC (QingDao) Co., Ltd.

Mr. Zeyong Luo Fuzhou Tianyu Electric Co., Ltd

Test location

Laboratory information

Testing laboratory Shandong Institute for Product Quality Inspection

Place of testing: No. 1, Chidong Avenue, Chiping District, Liaocheng, Shandong, P.R. China.

Testing period 2025-05-27

General remarks

The test results presented in this report relate only to the object tested.
This report shall not be reproduced, except in full, without the written approval of the Issuing testing laboratory.
"(See Enclosure document)" refers to additional information appended to the report.
"(See appended table)" refers to a table appended to the report.
Throughout this report a ☐ comma / ☒ point is used as the decimal separator.

Possible test case verdicts:

- test case does not apply to the test object : N/A
- test case for reference : R (Reference)
- test object does meet the requirement : P (Pass)
- test object does not meet the requirement : F (Fail)

Product information**Rated characteristics of the tested object assigned by clients**

Rating characteristics of the power Transformer	
Manufacturer	Fuzhou Tianyu Electric Co., Ltd
Designation	S-M-630/22-Tier2
Serial number/year of manufacture	BY04101/2025-05
Sampling procedure	By the manufacturer
Rated power (S_r)	630 kVA
Rated voltage (HV/LV) (U_r)	22($\pm 2 \times 2.5\%$)/0.42 kV
Rated frequency (f_r)	50 Hz
Rated current (HV/LV) (I_r)	16.53/866 A
Rated insulation level (Um/LI/LIC/AC) of HV winding	24/125/138/50 kV
Rated insulation level (Um/LI/LIC/AC) of LV winding	1.1/20/22/10 kV
Rated insulation level (Um/LI/AC) of LVN	1.1/20/10 kV
Cooling method	ONAN
Connection symbol	Dyn5
Insulation	Paper / Oil
Sound power level	≤ 45 dB(A)
Load losses at 75 °C at rated frequency of 50 Hz	≤ 4.6 kW
No-load current at rated voltage and frequency	0.4% $+30\%$
No-load losses at rated voltage and frequency	≤ 0.54 kW
Short-circuit impedance	6% ($\pm 10\%$)
Maximum level of losses for no-load losses and load losses	Class TIER2
Material of HV/LV windings	Copper
Nature of the core material	Silicon steel sheet
Insulating oil mass	495 kg
Total weight	2475 kg

Test report

Report Number: CN2532PX 001

Nameplate information

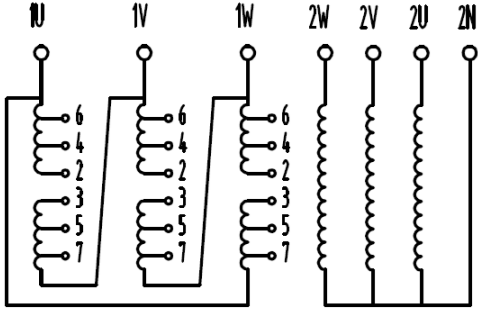


TRANSFORMER

PHASE : 3
RATED POWER : 630kVA
TYPE : S-M-630/22-Tier2
FREQUENCY : 50HZ
IMPEDANCE VOLT : 5.916 %
PRODUCT NO : 1RB.710.8827.1
STANDARD : MSZ EN 60076-1.2.3.5
EcoDesign-Tier 2
VECTOR GROUP SYMBOL : Dyn5
TYPE OF COOLING : ONAN
SERVICE CONDITION : OUT DOOR

TAPPING POSITION	HV		LV1	
	V	A	V	A
1	23100			
2	22550			
3	22000	16.53	420	866.0
4	21450			
5	20900			

INSULATION LEVEL : H.V. Um/LI/LIC/AC 24/125/138/50 kV
L.V. Um/LI/LIC/AC 11/20/22/10 kV



SERIAL NO: BY04101

1W	1V	1U	
○	○	○	
○	○	○	
2W	2V	2U	2N

SHIPPING : 2475 kg
BODY : 1765 kg
OIL : 495 kg
TOTAL : 2475 kg
NO PCB
COATING : EPOXY PRIMER POWDER WITH POLYURETHANE TOPCOAT POWDER
FUZHOU TIANYU ELECTRIC CO.,LTD

OIL NO. : I-20 ℃
OIL BASE : NAPHTHENE BASE
WIRE MATERIAL : Cu
PO : 0.526 kW
Pk : 4.472 kW
NOISE LEVEL : 45 dB
WARRANTY PERIOD : 2028/3/19
MADE IN 2025-05

Prüfbericht-Nr.: CN2532PX 001		Seite 8 von 28 Page 8 of 28	
Test report no.:			
Absatz Clause	Anforderungen - Prüfungen / Requirements – Tests EN 50708-2-1:2020	Messergebnisse – Bemerkungen/ Measuring results - Remarks	Ergebnis Result

1	Scope		P
	The scope of this document is to define the energy performance of Medium Power Transformers in compliance with EN 50708-1-1:2020.	Sr=630 kVA	P
	NOTE This document covers the transformers under Commission Regulation (EU) No 548/2014 of 21 May 2014 and its amendment No 2019/1783 of 1 October 2019, gives additional specific guidance for single phase transformers, autotransformers, multi winding transformers and for transformers with OD and OF cooling systems, necessary for the correct application of energy efficiency requirements to these categories of transformers.		P
	Medium power transformer means power transformer with all windings having a rated power lower than or equal to 3150 kVA, and highest voltage for equipment greater than 1.1 kV and lower than or equal to 36 kV.	Rated voltage (HV/LV) (Ur/Un): 22/0.42 kV	P
4	Service conditions		P
	See EN 60076-1:2011.		P
5	Rating and general requirements		P
5.1	General		P
5.1.1	Highest voltages for equipment for winding with $U_m > 1,1$ kV		P
	Insulation levels and dielectric test shall be in accordance with the requirements of EN 60076-3:2013 and for dry type transformers in accordance with EN IEC 60076-11:2018.	Oil-immersed power transformer comply with EN 60076-3:2013	P
	The values of the highest voltage U_m for equipment are: 3,6 kV – 7,2 kV – 12 kV – 17,5 kV – 24 kV – 36 kV	$U_m=24$ kV	P
	NOTE National practices might require the use of highest voltages for equipment up to (but not including) 52 kV, when the rated voltage is less than 36 kV (such as $U_m = 38,5$ kV or $U_m = 40,5$ kV).		N/A
5.1.2	Rated voltage for winding with $U_m \leq 1,1$ kV		P
	For $U_m \leq 1,1$ kV, the preferred rated voltage value shall be chosen in the hereunder list: 400 V – 410 V – 415 V – 420 V – 433 V – 690 V	$U_m=1.1$ kV	P

Prüfbericht-Nr.: CN2532PX 001 Test report no.:		Seite 9 von 28 Page 9 of 28	
Absatz Clause	Anforderungen - Prüfungen / Requirements – Tests EN 50708-2-1:2020	Messergebnisse – Bemerkungen/ Measuring results - Remarks	Ergebnis Result
	This document may be applied either as a whole or in part, to transformers with rated low voltages below 400 V and above 690 V.		N/A
	The values for short duration power frequency withstand and/or for the lightning impulse withstand test between windings and earth need to be stated by the purchaser.		N/A
5.1.3	Tapping		P
	For DETC (De-energized tap changer), the preferred tapping ranges shall be $\pm 2,5 \%$ with 3 tap positions and $\pm 2 \times 2,5 \%$ with 5 tap positions. On special request $\pm 4 \times 2,5 \%$ with 9 tap positions can be provided. Tapping ranges greater than $\pm 10 \%$ or with more than 9 tap positions are unusual and subject to specific agreement.	DETC (De-energized tap changer) Tapping range: $\pm 2 \times 2.5 \%$	P
	Tapping ranges outside the above definitions for DETC or any ranges for OLTC (On load tap changer) shall be specified by agreement between manufacturer and purchaser.		N/A
	The level of losses of this standard shall be compliant whatever the ranges and methods of tapping.		P
	NOTE OLTC means also transformers equipped with tapping connections suitable for operation while transformer is energised or on load for voltage adaptation purposes.		N/A
5.1.4	Connection designations for three phase transformers		P
	Preferred connection designations (IEV [421-10-09]) shall be in accordance with clock notation 5 or 11 with Dyn, unless otherwise agreed between manufacturer and purchaser.	Dyn5	P
5.2	Energy Performance		P
5.2.1	General		P
5.2.1.1	Introduction		P
	The tables below indicate the maximum level of losses for no-load losses and for load losses for TIER1 and TIER2.	This oil-immersed power transformer complied with TIER2	P
	TIER1 has been applied since 1 July 2015 for the values of losses following Commission Regulation (EU) No 548/2014 of 21 May 2014 and its amendment No 2019/1783 of 1 October 2019.		N/A

Prüfbericht-Nr.: CN2532PX 001 Test report no.:		Seite 10 von 28 Page 10 of 28	
Absatz Clause	Anforderungen - Prüfungen / Requirements – Tests EN 50708-2-1:2020	Messergebnisse – Bemerkungen/ Measuring results - Remarks	Ergebnis Result
	TIER2 shall be applied from 1 July 2021 for the values of losses following Commission Regulation (EU) No 548/2014 of 21 May 2014 and its amendment No 2019/1783 of 1 October 2019.		P
	Table 1 and 2 define load losses and no-load losses for one winding with $U_m \leq 24$ kV and the other one with $U_m \leq 3,6$ kV. Additional corrections are defined for highest voltage, dual voltage or special on load tapping, given in Table 3, 4, 5 and other clauses. Corrections factors in Table 3, 4, 5 are cumulative.		P
	During factory acceptance tests, the measured values of no-load and load losses shall not exceed the respective maximum values determined with Table 1 and 2 and 6 and corrected by corrections factor as stated above.		P
	The reference conditions defined in EN 50708-1-1:2020 applies.		P
	Other values of short-circuit impedance can be specified by the purchaser for particular system service conditions, e.g. in the case of parallel operations in all tables of this clause.		N/A
	The sound power levels are given in Tables 1 and 2. Transformers can be requested or offered with sound power level L_{WA} other than the listed values.	Sound power level: ≤ 45 dB(A)	P
	Unless otherwise agreed between manufacturer and purchaser the sound power level L_{WA} given in Tables 1 and 2 are considered for no load condition.		N/A
	NOTE Loss levels and correction factors defined in 5.2.1.3 and 5.2.1.4 refer to two winding medium power transformers. Loss levels for three windings medium power transformers are not defined due to lack of data at the time of the publication of this standard.		P
	EXAMPLE cumulative corrections:		N/A
	In case of transformers having primary highest voltage for equipment $U_m = 36$ kV and secondary highest voltage for equipment $U_m \leq 3,6$ kV allowing 10 % on P_k and 15 % on P_0 according to Table 3, and having dual voltage on one winding allowing 10 % on P_k and 15 % on P_0 according to Table 4.		N/A
	The total corrections on P_k is 21 % (1,1x1,1–1) and P_0 is 32,25 % (1,15x1,15–1)		N/A

Prüfbericht-Nr.: CN2532PX 001		Seite 11 von 28 Page 11 of 28																																																																																																																															
Test report no.:																																																																																																																																	
Absatz Clause	Anforderungen - Prüfungen / Requirements – Tests EN 50708-2-1:2020	Messergebnisse – Bemerkungen/ Measuring results - Remarks	Ergebnis Result																																																																																																																														
5.2.1.2	Load losses and no-load losses for one winding with $U_m \leq 24$ kV and the other one with $U_m \leq 3,6$ kV		P																																																																																																																														
	The following tables are defined for step down or step up transformers with one winding with $U_m \leq 24$ kV and the other one with $U_m \leq 3,6$ kV. Table 1 is for liquid immersed transformers and Table 2 for dry-type transformers.	Oil-immersed power transformer	P																																																																																																																														
	Table 1 — Losses, short circuit impedance and sound power level for liquid immersed transformers <table><tr><th>Rated power</th><th>TIER1 P_k</th><th>TIER1 P_0</th><th>TIER2 P_k</th><th>TIER2 P_0</th><th>L_{WA}</th><th>Short-circuit Impedance</th></tr><tr><th>kVA</th><th>W</th><th>W</th><th>W</th><th>W</th><th>dB(A)</th><th>%</th></tr><tr><td>≤ 25</td><td>900</td><td>70</td><td>600</td><td>63</td><td>37</td><td>4</td></tr><tr><td>50</td><td>1 100</td><td>90</td><td>750</td><td>81</td><td>39</td><td>4</td></tr><tr><td>100</td><td>1 750</td><td>145</td><td>1 250</td><td>130</td><td>41</td><td>4</td></tr><tr><td>160</td><td>2 350</td><td>210</td><td>1 750</td><td>189</td><td>44</td><td>4</td></tr><tr><td>250</td><td>3 250</td><td>300</td><td>2 350</td><td>270</td><td>47</td><td>4</td></tr><tr><td>315</td><td>3 900</td><td>360</td><td>2 800</td><td>324</td><td>49</td><td>4</td></tr><tr><td>400</td><td>4 600</td><td>430</td><td>3 250</td><td>387</td><td>50</td><td>4</td></tr><tr><td>500</td><td>5 500</td><td>510</td><td>3 900</td><td>459</td><td>51</td><td>4</td></tr><tr><td>630</td><td>6 500</td><td>600</td><td>4 600</td><td>540</td><td>52</td><td>4 or 6</td></tr><tr><td>800</td><td>8 400</td><td>650</td><td>6 000</td><td>585</td><td>53</td><td>6</td></tr><tr><td>1 000</td><td>10 500</td><td>770</td><td>7 600</td><td>693</td><td>55</td><td>6</td></tr><tr><td>1 250</td><td>11 000</td><td>950</td><td>9 500</td><td>855</td><td>56</td><td>6</td></tr><tr><td>1 600</td><td>14 000</td><td>1 200</td><td>12 000</td><td>1 080</td><td>58</td><td>6</td></tr><tr><td>2 000</td><td>18 000</td><td>1 450</td><td>15 000</td><td>1 305</td><td>60</td><td>6</td></tr><tr><td>2 500</td><td>22 000</td><td>1 750</td><td>18 500</td><td>1 575</td><td>63</td><td>6</td></tr><tr><td>3 150</td><td>27 500</td><td>2 200</td><td>23 000</td><td>1 980</td><td>64</td><td>6</td></tr></table>	Rated power	TIER1 P_k	TIER1 P_0	TIER2 P_k	TIER2 P_0	L_{WA}	Short-circuit Impedance	kVA	W	W	W	W	dB(A)	%	≤ 25	900	70	600	63	37	4	50	1 100	90	750	81	39	4	100	1 750	145	1 250	130	41	4	160	2 350	210	1 750	189	44	4	250	3 250	300	2 350	270	47	4	315	3 900	360	2 800	324	49	4	400	4 600	430	3 250	387	50	4	500	5 500	510	3 900	459	51	4	630	6 500	600	4 600	540	52	4 or 6	800	8 400	650	6 000	585	53	6	1 000	10 500	770	7 600	693	55	6	1 250	11 000	950	9 500	855	56	6	1 600	14 000	1 200	12 000	1 080	58	6	2 000	18 000	1 450	15 000	1 305	60	6	2 500	22 000	1 750	18 500	1 575	63	6	3 150	27 500	2 200	23 000	1 980	64	6	TIER2 Sr=630 kVA Pk≤4600 W P0≤540 W LWA≤45 dB(A)	P
Rated power	TIER1 P_k	TIER1 P_0	TIER2 P_k	TIER2 P_0	L_{WA}	Short-circuit Impedance																																																																																																																											
kVA	W	W	W	W	dB(A)	%																																																																																																																											
≤ 25	900	70	600	63	37	4																																																																																																																											
50	1 100	90	750	81	39	4																																																																																																																											
100	1 750	145	1 250	130	41	4																																																																																																																											
160	2 350	210	1 750	189	44	4																																																																																																																											
250	3 250	300	2 350	270	47	4																																																																																																																											
315	3 900	360	2 800	324	49	4																																																																																																																											
400	4 600	430	3 250	387	50	4																																																																																																																											
500	5 500	510	3 900	459	51	4																																																																																																																											
630	6 500	600	4 600	540	52	4 or 6																																																																																																																											
800	8 400	650	6 000	585	53	6																																																																																																																											
1 000	10 500	770	7 600	693	55	6																																																																																																																											
1 250	11 000	950	9 500	855	56	6																																																																																																																											
1 600	14 000	1 200	12 000	1 080	58	6																																																																																																																											
2 000	18 000	1 450	15 000	1 305	60	6																																																																																																																											
2 500	22 000	1 750	18 500	1 575	63	6																																																																																																																											
3 150	27 500	2 200	23 000	1 980	64	6																																																																																																																											
	Table 2 — Losses, short circuit impedance and sound power levels for dry-type transformers <table><tr><th>Rated power</th><th>TIER1 P_k</th><th>TIER1 P_0</th><th>TIER2 P_k</th><th>TIER2 P_0</th><th>L_{WA}</th><th>Short-circuit Impedance</th></tr><tr><th>kVA</th><th>W</th><th>W</th><th>W</th><th>W</th><th>dB(A)</th><th>%</th></tr><tr><td>≤ 50</td><td>1 700</td><td>200</td><td>1 500</td><td>180</td><td>49</td><td>6</td></tr><tr><td>100</td><td>2 050</td><td>280</td><td>1 800</td><td>252</td><td>51</td><td>6</td></tr><tr><td>160</td><td>2 900</td><td>400</td><td>2 600</td><td>360</td><td>54</td><td>6</td></tr><tr><td>250</td><td>3 800</td><td>520</td><td>3 400</td><td>468</td><td>57</td><td>6</td></tr><tr><td>400</td><td>5 500</td><td>750</td><td>4 500</td><td>675</td><td>60</td><td>6</td></tr><tr><td>630</td><td>7 600</td><td>1 100</td><td>7 100</td><td>990</td><td>62</td><td>6</td></tr><tr><td>800</td><td>8 000</td><td>1 300</td><td>8 000</td><td>1 170</td><td>64</td><td>6</td></tr><tr><td>1 000</td><td>9 000</td><td>1 550</td><td>9 000</td><td>1 395</td><td>65</td><td>6</td></tr><tr><td>1 250</td><td>11 000</td><td>1 800</td><td>11 000</td><td>1 620</td><td>67</td><td>6</td></tr><tr><td>1 600</td><td>13 000</td><td>2 200</td><td>13 000</td><td>1 980</td><td>68</td><td>6</td></tr><tr><td>2 000</td><td>16 000</td><td>2 600</td><td>16 000</td><td>2 340</td><td>70</td><td>6</td></tr><tr><td>2 500</td><td>19 000</td><td>3 100</td><td>19 000</td><td>2 790</td><td>71</td><td>6</td></tr><tr><td>3 150</td><td>22 000</td><td>3 800</td><td>22 000</td><td>3 420</td><td>74</td><td>6</td></tr></table>	Rated power	TIER1 P_k	TIER1 P_0	TIER2 P_k	TIER2 P_0	L_{WA}	Short-circuit Impedance	kVA	W	W	W	W	dB(A)	%	≤ 50	1 700	200	1 500	180	49	6	100	2 050	280	1 800	252	51	6	160	2 900	400	2 600	360	54	6	250	3 800	520	3 400	468	57	6	400	5 500	750	4 500	675	60	6	630	7 600	1 100	7 100	990	62	6	800	8 000	1 300	8 000	1 170	64	6	1 000	9 000	1 550	9 000	1 395	65	6	1 250	11 000	1 800	11 000	1 620	67	6	1 600	13 000	2 200	13 000	1 980	68	6	2 000	16 000	2 600	16 000	2 340	70	6	2 500	19 000	3 100	19 000	2 790	71	6	3 150	22 000	3 800	22 000	3 420	74	6		N/A																					
Rated power	TIER1 P_k	TIER1 P_0	TIER2 P_k	TIER2 P_0	L_{WA}	Short-circuit Impedance																																																																																																																											
kVA	W	W	W	W	dB(A)	%																																																																																																																											
≤ 50	1 700	200	1 500	180	49	6																																																																																																																											
100	2 050	280	1 800	252	51	6																																																																																																																											
160	2 900	400	2 600	360	54	6																																																																																																																											
250	3 800	520	3 400	468	57	6																																																																																																																											
400	5 500	750	4 500	675	60	6																																																																																																																											
630	7 600	1 100	7 100	990	62	6																																																																																																																											
800	8 000	1 300	8 000	1 170	64	6																																																																																																																											
1 000	9 000	1 550	9 000	1 395	65	6																																																																																																																											
1 250	11 000	1 800	11 000	1 620	67	6																																																																																																																											
1 600	13 000	2 200	13 000	1 980	68	6																																																																																																																											
2 000	16 000	2 600	16 000	2 340	70	6																																																																																																																											
2 500	19 000	3 100	19 000	2 790	71	6																																																																																																																											
3 150	22 000	3 800	22 000	3 420	74	6																																																																																																																											
5.2.1.3	Load loss and no-load loss in case of other highest voltage		N/A																																																																																																																														
	The level of losses given in Table 1 and in Table 2 shall be weighted by the correction factors given in Table 3 below, in order to take care of variations related to the highest voltage for equipment values.		N/A																																																																																																																														

Prüfbericht-Nr.: CN2532PX 001

Test report no.:

Seite 12 von 28
Page 12 of 28

Absatz Clause	Anforderungen - Prüfungen / Requirements – Tests EN 50708-2-1:2020	Messergebnisse – Bemerkungen/ Measuring results - Remarks	Ergebnis Result																																				
	Table 3— Corrections of the losses for highest voltage <table border="1"> <thead> <tr> <th colspan="2">Special combination of voltages in one winding</th><th>Load losses (P_k)</th><th>No-load losses (P₀)</th></tr> </thead> <tbody> <tr> <td colspan="4">For both liquid immersed (Table 1) and dry type (Table 2)</td></tr> <tr> <td>Primary highest voltage for equipment U_m ≤ 24kV</td><td>Secondary highest voltage for equipment U_m > 3,6kV</td><td>No correction</td><td>No correction</td></tr> <tr> <td colspan="4">For liquid immersed (Table 1)</td></tr> <tr> <td>Primary highest voltage for equipment U_m = 36kV</td><td>Secondary highest voltage for equipment U_m ≤ 3,6kV</td><td>10 %</td><td>15 %</td></tr> <tr> <td>Primary highest voltage for equipment U_m = 36kV</td><td>Secondary highest voltage for equipment U_m > 3,6kV</td><td>10 %</td><td>15 %</td></tr> <tr> <td colspan="4">For dry type (Table 2)</td></tr> <tr> <td>Primary highest voltage for equipment U_m = 36kV</td><td>Secondary highest voltage for equipment U_m ≤ 3,6kV</td><td>10 %</td><td>15 %</td></tr> <tr> <td>Primary highest voltage for equipment U_m = 36kV</td><td>Secondary highest voltage for equipment U_m > 3,6kV</td><td>15 %</td><td>20 %</td></tr> </tbody> </table>	Special combination of voltages in one winding		Load losses (P _k)	No-load losses (P ₀)	For both liquid immersed (Table 1) and dry type (Table 2)				Primary highest voltage for equipment U _m ≤ 24kV	Secondary highest voltage for equipment U _m > 3,6kV	No correction	No correction	For liquid immersed (Table 1)				Primary highest voltage for equipment U _m = 36kV	Secondary highest voltage for equipment U _m ≤ 3,6kV	10 %	15 %	Primary highest voltage for equipment U _m = 36kV	Secondary highest voltage for equipment U _m > 3,6kV	10 %	15 %	For dry type (Table 2)				Primary highest voltage for equipment U _m = 36kV	Secondary highest voltage for equipment U _m ≤ 3,6kV	10 %	15 %	Primary highest voltage for equipment U _m = 36kV	Secondary highest voltage for equipment U _m > 3,6kV	15 %	20 %		N/A
Special combination of voltages in one winding		Load losses (P _k)	No-load losses (P ₀)																																				
For both liquid immersed (Table 1) and dry type (Table 2)																																							
Primary highest voltage for equipment U _m ≤ 24kV	Secondary highest voltage for equipment U _m > 3,6kV	No correction	No correction																																				
For liquid immersed (Table 1)																																							
Primary highest voltage for equipment U _m = 36kV	Secondary highest voltage for equipment U _m ≤ 3,6kV	10 %	15 %																																				
Primary highest voltage for equipment U _m = 36kV	Secondary highest voltage for equipment U _m > 3,6kV	10 %	15 %																																				
For dry type (Table 2)																																							
Primary highest voltage for equipment U _m = 36kV	Secondary highest voltage for equipment U _m ≤ 3,6kV	10 %	15 %																																				
Primary highest voltage for equipment U _m = 36kV	Secondary highest voltage for equipment U _m > 3,6kV	15 %	20 %																																				
5.2.1.4	Load loss and no-load loss in case of dual voltage		N/A																																				
	Table 4 below gives correction factors to be applied to the load and no-load losses indicated in Tables 1, 2 and 6 for medium power transformers with dual voltage in one or both windings differing more than 10 %.		N/A																																				
	The losses shall be calculated on the base of the voltage of the winding specified in the second column of Table 4 and can be increased with the correction factors given in the last 2 columns of Table 4. In any case, whatever the combinations of winding voltages, the losses cannot exceed the values given in Tables 1, 2 and 6 corrected by the factors in Table 4.		N/A																																				
	Table 4— Corrections of the losses for dual voltage <table border="1"> <thead> <tr> <th>Type of dual voltage</th><th>Reference voltage for the application of correction factors</th><th>Load losses (P_k)</th><th>No-load losses (P₀)</th></tr> </thead> <tbody> <tr> <td>Dual voltage on one winding with reduced power output on the lower low-voltage winding AND maximum available power on the lower voltage of the low-voltage winding limited to 0,85 of the rated power assigned to the low-voltage winding at its higher voltage.</td><td>Losses shall be calculated based on the higher voltage of the low-voltage winding</td><td>No correction</td><td>No correction</td></tr> <tr> <td>Dual voltage on one winding with reduced power output on the lower high-voltage winding AND maximum available power on the lower voltage of the high-voltage winding limited to 0,85 of the rated power assigned to the high-voltage winding at its higher voltage.</td><td>Losses shall be calculated based on the higher voltage of the high-voltage winding</td><td>No correction</td><td>No correction</td></tr> <tr> <td>Dual voltage on one winding AND full rated power available on both windings, i.e. the full nominal power is available regardless of the combination of voltages.</td><td>The losses shall be calculated based on the higher voltage of the dual voltage winding</td><td>10 %</td><td>15 %</td></tr> <tr> <td>Dual voltage on both windings AND rated power available on all combinations of windings, i.e. both voltages on one winding are fully rated in combination with one of the voltages on the other winding</td><td>The losses shall be calculated based on the higher voltages of both dual voltage windings</td><td>20 %</td><td>20 %</td></tr> </tbody> </table>	Type of dual voltage	Reference voltage for the application of correction factors	Load losses (P _k)	No-load losses (P ₀)	Dual voltage on one winding with reduced power output on the lower low-voltage winding AND maximum available power on the lower voltage of the low-voltage winding limited to 0,85 of the rated power assigned to the low-voltage winding at its higher voltage.	Losses shall be calculated based on the higher voltage of the low-voltage winding	No correction	No correction	Dual voltage on one winding with reduced power output on the lower high-voltage winding AND maximum available power on the lower voltage of the high-voltage winding limited to 0,85 of the rated power assigned to the high-voltage winding at its higher voltage.	Losses shall be calculated based on the higher voltage of the high-voltage winding	No correction	No correction	Dual voltage on one winding AND full rated power available on both windings, i.e. the full nominal power is available regardless of the combination of voltages.	The losses shall be calculated based on the higher voltage of the dual voltage winding	10 %	15 %	Dual voltage on both windings AND rated power available on all combinations of windings, i.e. both voltages on one winding are fully rated in combination with one of the voltages on the other winding	The losses shall be calculated based on the higher voltages of both dual voltage windings	20 %	20 %		N/A																
Type of dual voltage	Reference voltage for the application of correction factors	Load losses (P _k)	No-load losses (P ₀)																																				
Dual voltage on one winding with reduced power output on the lower low-voltage winding AND maximum available power on the lower voltage of the low-voltage winding limited to 0,85 of the rated power assigned to the low-voltage winding at its higher voltage.	Losses shall be calculated based on the higher voltage of the low-voltage winding	No correction	No correction																																				
Dual voltage on one winding with reduced power output on the lower high-voltage winding AND maximum available power on the lower voltage of the high-voltage winding limited to 0,85 of the rated power assigned to the high-voltage winding at its higher voltage.	Losses shall be calculated based on the higher voltage of the high-voltage winding	No correction	No correction																																				
Dual voltage on one winding AND full rated power available on both windings, i.e. the full nominal power is available regardless of the combination of voltages.	The losses shall be calculated based on the higher voltage of the dual voltage winding	10 %	15 %																																				
Dual voltage on both windings AND rated power available on all combinations of windings, i.e. both voltages on one winding are fully rated in combination with one of the voltages on the other winding	The losses shall be calculated based on the higher voltages of both dual voltage windings	20 %	20 %																																				

Prüfbericht-Nr.: Test report no.:		CN2532PX 001		Seite 13 von 28 Page 13 of 28										
Absatz Clause	Anforderungen - Prüfungen / Requirements – Tests EN 50708-2-1:2020			Messergebnisse – Bemerkungen/ Measuring results - Remarks	Ergebnis Result									
	In case of transformers with two identical low voltage windings and/or high voltage windings in parallel connection designed with half of rated power each, dual voltage correction factors given in Table 4 should be applied.				N/A									
	NOTE In case of transformers with dual high voltage winding, the rated power considered is on the highest rated voltage winding.				N/A									
5.2.2	Transformers with special requirements				N/A									
5.2.2.1	Rated power not included in the list				N/A									
	For transformer having a rated power different from the values given in the tables, a linear interpolation shall be done.				N/A									
5.2.2.2	Transformers equipped for voltage regulation while energised				N/A									
	When transformers are equipped with tapping connections suitable for operation while transformer is energised or on load for voltage adaptation purposes, the level of load loss and no-load loss of the list of this standard shall be increased.				N/A									
	Table 5—Correction of losses for transformers equipped for voltage regulation <table><tr><td></td><td>P_k</td><td>P₀</td></tr><tr><td>TIER1</td><td>5 %</td><td>20 %</td></tr><tr><td>TIER2</td><td>No correction</td><td>10 %</td></tr></table>				P _k	P ₀	TIER1	5 %	20 %	TIER2	No correction	10 %		N/A
	P _k	P ₀												
TIER1	5 %	20 %												
TIER2	No correction	10 %												
	Such corrections are applicable irrespectively of the technology used to perform the voltage adaptation.				N/A									
	The losses of the transformers are those measured during the test whatever the kind of transformers, included all devices to manage the AVR inside the transformers.				N/A									
	For a transformer having an insulation level according to Table 3 and having tapping connections suitable for operation while the transformer is energized or on load for voltage adaptation purposes both corrections shall be taken into account.				N/A									
	NOTE AVR: Automatic Voltage Regulation: to automatically control the input or output voltage of the transformer for on-load voltage regulation purposes.				N/A									
5.2.2.3	Liquid immersed three phase pole mounted transformers				N/A									

Prüfbericht-Nr.: Test report no.:		CN2532PX 001		Seite 14 von 28 Page 14 of 28																																																			
Absatz Clause	Anforderungen - Prüfungen / Requirements – Tests EN 50708-2-1:2020			Messergebnisse – Bemerkungen/ Measuring results - Remarks	Ergebnis Result																																																		
	For the one-to-one replacement of existing medium power pole-mounted transformers the applicable maximum levels of load and no-load losses are not the ones in Tables 1 and 2 but those in Table 6 below. Maximum allowable losses for kVA ratings other than those explicitly mentioned in Table 6 shall be obtained by linear interpolation or extrapolation. The correction factors for special combinations of winding voltages indicated in Tables 3 and 4 are also applicable.				N/A																																																		
	For the one-to-one replacement of existing medium power pole-mounted transformers, the manufacturer, importer or authorized representative shall include in the technical documentation of the transformer the following information: — the address and contact details of the commissioner of the replacement transformer; — the station where the replacement transformer is to be installed. This shall be unequivocally identified either by a specific location or a specific installation type (e.g. technical description of the pole).				N/A																																																		
	In the above cases, the manufacturer, importer or authorized representative shall notify the competent national market surveillance authorities.				N/A																																																		
	With regards to the installation of new pole-mounted transformers, it is the requirements in Tables 1 and 2, in conjunction with Tables 3 and 4 where justified, which are applicable.				N/A																																																		
	Table 6 — Maximum load and no-load losses for medium power immersed pole mounted transformers for one to one replacement <table><tr><td>Rated power</td><td>TIER1 P_k</td><td>TIER1 P₀</td><td>TIER2 P_k</td><td>TIER2 P₀</td></tr><tr><td>kVA</td><td>W</td><td>W</td><td>W</td><td>W</td></tr><tr><td>25</td><td>900</td><td>70</td><td>725</td><td>70</td></tr><tr><td>50</td><td>1 100</td><td>90</td><td>875</td><td>90</td></tr><tr><td>100</td><td>1 750</td><td>145</td><td>1 475</td><td>145</td></tr><tr><td>160</td><td>3 102</td><td>300</td><td>3 102</td><td>270</td></tr><tr><td>200</td><td>2 750</td><td>356</td><td>2 333</td><td>310</td></tr><tr><td>250</td><td>3 250</td><td>425</td><td>2 750</td><td>360</td></tr><tr><td>315</td><td>3 900</td><td>520</td><td>3 250</td><td>440</td></tr><tr><td>400</td><td>4 750</td><td>645</td><td>3 905</td><td>545</td></tr></table>			Rated power	TIER1 P _k	TIER1 P ₀	TIER2 P _k	TIER2 P ₀	kVA	W	W	W	W	25	900	70	725	70	50	1 100	90	875	90	100	1 750	145	1 475	145	160	3 102	300	3 102	270	200	2 750	356	2 333	310	250	3 250	425	2 750	360	315	3 900	520	3 250	440	400	4 750	645	3 905	545		N/A
Rated power	TIER1 P _k	TIER1 P ₀	TIER2 P _k	TIER2 P ₀																																																			
kVA	W	W	W	W																																																			
25	900	70	725	70																																																			
50	1 100	90	875	90																																																			
100	1 750	145	1 475	145																																																			
160	3 102	300	3 102	270																																																			
200	2 750	356	2 333	310																																																			
250	3 250	425	2 750	360																																																			
315	3 900	520	3 250	440																																																			
400	4 750	645	3 905	545																																																			
6	Rating plate				P																																																		
	The additional requirements for rating plate are given in EN 50708-1-1:2020.				P																																																		

Prüfbericht-Nr.: CN2532PX 001 Test report no.:		Seite 15 von 28 Page 15 of 28	
Absatz Clause	Anforderungen - Prüfungen / Requirements – Tests EN 50708-2-1:2020	Messergebnisse – Bemerkungen/ Measuring results - Remarks	Ergebnis Result
	The one-to-one replacement of existing medium power pole-mounted transformers shall be written on the rating plate by the words "1/1 replacement"		P
7	Tolerances during factory acceptance tests		P
	The additional requirements for tolerances are given in EN 50708-1-1:2020.		P
	During factory acceptance tests, the measured values of no load and load losses shall not exceed the respective maximum values specified in EN 50708-2-1:2020 without tolerances, and the Peak Efficiency Index values obtained from the measured values of no load and load losses shall not be below the respective minimum.		P
	For other test tolerances see EN 60076 series of standards.		P
	During market surveillance measurements, the individual measured values of loss can be up to 5 % higher than the declared or calculated ones, in the case of PEI values.		N/A
	During market surveillance measurements, the measured values of electrical power required by the cooling system can be up to 5 % higher than the declared value of electrical power required by the cooling system.		N/A
8	Tests		N/A
	The additional requirements for tests are given in EN 50708-1-1:2020 or in relevant parts.		N/A
9	Accessories and fittings		N/A
	List of accessories and fittings are under development in relevant parts.		N/A
10	Capitalization losses		N/A
	The capitalization should be applied to fixed losses by choosing the adequate rated power or by specifying lower losses than the maximum values given in the Clause 5.		N/A
11	Transformers overhaul		N/A
	The additional requirements for transformers overhaul are given in EN 50708-1-1:2020.		N/A

Prüfbericht-Nr.: <i>Test report no.:</i>	CN2532PX 001	Seite 16 von 28 Page 16 of 28	
Absatz <i>Clause</i>	Anforderungen - Prüfungen / <i>Requirements – Tests</i> EN 50708-2-1:2020	Messergebnisse – Bemerkungen/ <i>Measuring results - Remarks</i>	Ergebnis <i>Result</i>
Annex ZZ	Relationship between this European Standard and the ecodesign requirements of Commission Regulation (EU) No 548/2014 of 21 May 2014 and its amendment No 2019/1783 of 1 October 2019 on implementing Directive 2009/125/EC of the European Parliament and of the Council with regard to small, medium and large power transformers aimed to be covered (informative)		P

Energy efficiency measurement and calculation

- ☒ Transformers with rated power $S_r \leq 3150$ kVA
Maximum level of losses are required in EN 50708-2-1:2020
Rated Power S_r : 630 kVA
Maximum level of losses for no-load losses and load losses: Class TIER2
No-load losses at rated voltage and frequency: ≤ 540 W
Sound power level: ≤ 45 dB(A)
Load losses at 75 °C at rated frequency of 50 Hz: ≤ 4600 W
Short circuit impedance between HV and LV in reference to the rated power: 6.0% ($\pm 10\%$)

Remarks: Detailed test items referred to Appendage A in this report.

Appendage A

A.1 Measurement of short-circuit impedance and load loss

Standard and date

Standard

Subclause 11.4 of EN 60076-1

Test date

2025-05-27

Environmental conditions

Ambient temperature

28.7 °C

Ambient air pressure

101.4 kPa

Relative humidity

27%

Characteristic test data

Serial number of power transformer

BY04101

Material of windings

Copper

Requirement

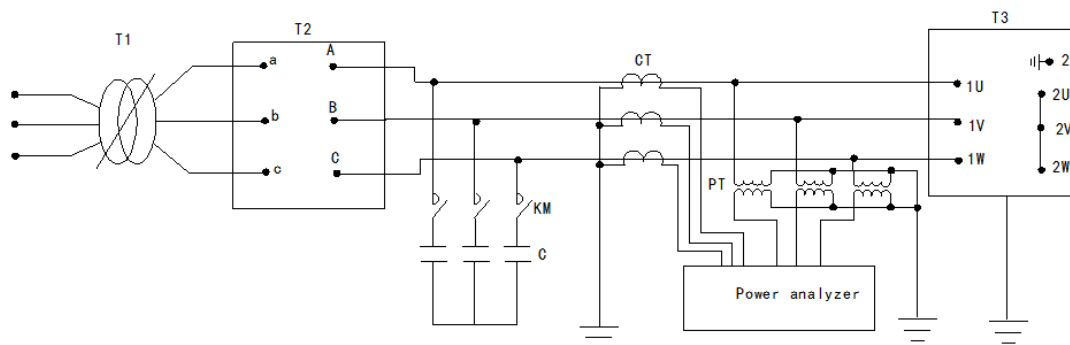
Load loss in tap 3 HV-LV combination

≤4.6 kW

Impedance voltage of HV-LV combination in tap 3

6.0% (1±10%)

Short-circuit impedance and load loss test circuit



T1	Voltage regulator	T2	Setp-up transformer
T3	Transformer tested	CT	Current transformer
PT	Voltage transformer	KM	Vacuum contactor
C	Capacitor		
Short-circuit impedance and load loss test system			

Procedure

The short-circuit impedance and load losses for a pair of windings were measured at rated frequency with voltage applied to the terminals of one winding, while the other winding was shorted.

The average oil temperature was taken as the mean of the top and bottom oil temperatures.

The measurements were taken as follows:

The combination HV at tapping position 1, 3, 5 with LV terminals shorted and current based on rated power.

The measuring results were recalculated for 75 °C winding temperature.

Test data

Measurement of short-circuit impedance and load loss									
HV tapping position	1			3			5		
Rated voltage of HV side [kV]	23.1			22.0			20.9		
Rated voltage of LV side [kV]	0.42			0.42			0.42		
Rated current of HV side I _{HVrated} [A]	15.746			16.533			17.403		
Rated current of LV side I _{LVrated} [A]	866.03			866.03			866.03		
Resistance at T _m 26.5 °C									
Measured value for HV winding [Ω]	1U1V	1V1W	1U1W	1U1V	1V1W	1U1W	1U1V	1V1W	1U1W
	4.895	4.911	4.910	4.635	4.648	4.646	4.374	4.388	4.387
Average value for HV winding (Phase to phase) R _{HV} [Ω]	4.905			4.643			4.383		
Measured value for LV winding [mΩ]	2U-2V			2V-2W			2U-2W		
	0.0013575			0.0013517			0.0013703		
Average value for LV winding (Phase to phase) R _{LV} [mΩ]	0.0013598								
Ohmic loss at rated power									
Calculated value at 28.7 °C P _{ct} [W]	3353.948554			3433.50093			3521.044316		
Correction value at 75 °C P _{ohmic loss} [W]	3976.000198			4070.30703			4174.086952		
Calculated equation: a. R _{θ1} = R _{TM} (225+θ ₁)/(225+T _m) b. P _{ct} = ΣI ² R _{θ1} = 1.5×I _{HV} ² R _{HVθ1} +1.5×I _{LV} ² R _{LVθ1} c. K = (225+75)/(225+θ ₁) d. P _{ohmic loss} = P _{ct} ×K									
Load loss at rated power (Liquid average temperature θ ₁ : 28.7 °C)									
Measured value at 28.7 °C P _{mt} [W]	2165.3			2447.2			2389.4		
Corrected measuring value (rated current) at 28.7 °C P _{cmt} [W]	3720.069093			3891.457679			3868.783594		
Additional loss at 28.7 °C P _{a2} [W]	366.1205395			457.9567496			347.7392775		
Correction additional loss at 75 °C P _{additional loss} [W]	311.4386654			389.5586931			295.8027338		
Load loss at 75 °C P _{load loss} [W]	4287.4			4459.9			4469.9		
Calculated equation: a. P _{cmt} = P _{mt} ×(I _{HVrated} /I _{ac}) ² b. P _{a2} = P _{cmt} -P _{ct} c. P _{additional loss} = P _{a2} ×(225+θ ₁)/(225+75) d. P _{load loss} = P _{ohmic loss} +P _{additional loss}									
Impedance voltage									
Applied current I _{ac} [A]	12.013			13.111			13.677		
Measured voltage [V]	1058.7			1026.8			948.2		
Short-circuit impedance	[%]	6.018			5.8957			5.7839	
	[Ω]	50.962			45.294			40.103	

Result

Passed, load loss in tap 3 HV-LV combination was less than the requirement of 4.6 kW. Impedance voltage of HV-LV combination in tap 3 met the requirement of 6.0% (1±10%).

Test report

Report Number: CN2532PX 001

Test photograph



A.2 Measurement of no-load loss and current

Standard and date

Standard

Subclause 11.5 of EN 60076-1

Test date

2025-05-27

Environmental conditions

Ambient temperature

28.7 °C

Ambient air pressure

101.5 kPa

Relative humidity

27%

Characteristic test data

Serial number of power transformer

BY04101

Material of windings

Copper

Applied rated voltage from LV windings

420 V

Requirement

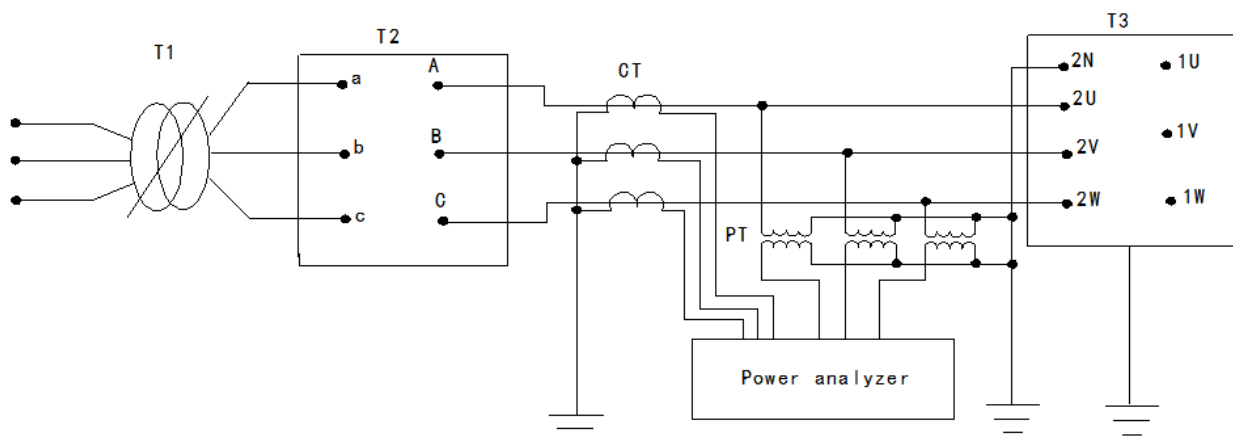
Maximum no-load loss at 100% voltage

≤0.54 kW

No-load current ratio

0.40%^{+30%}

No-load loss and current test circuit



T1	Voltage regulator	T2	Setp-up transformer
T3	Transformer tested	CT	Current transformer
PT	Voltage transformer		
No load loss and current test system			

Procedure

The no-load loss and no-load current were measured at 90%, 100% and 110% of the rated voltage at principal tapping (tap position 3) for HV winding, supplied to the low voltage terminals with 400 V and rated frequency of 50 Hz.

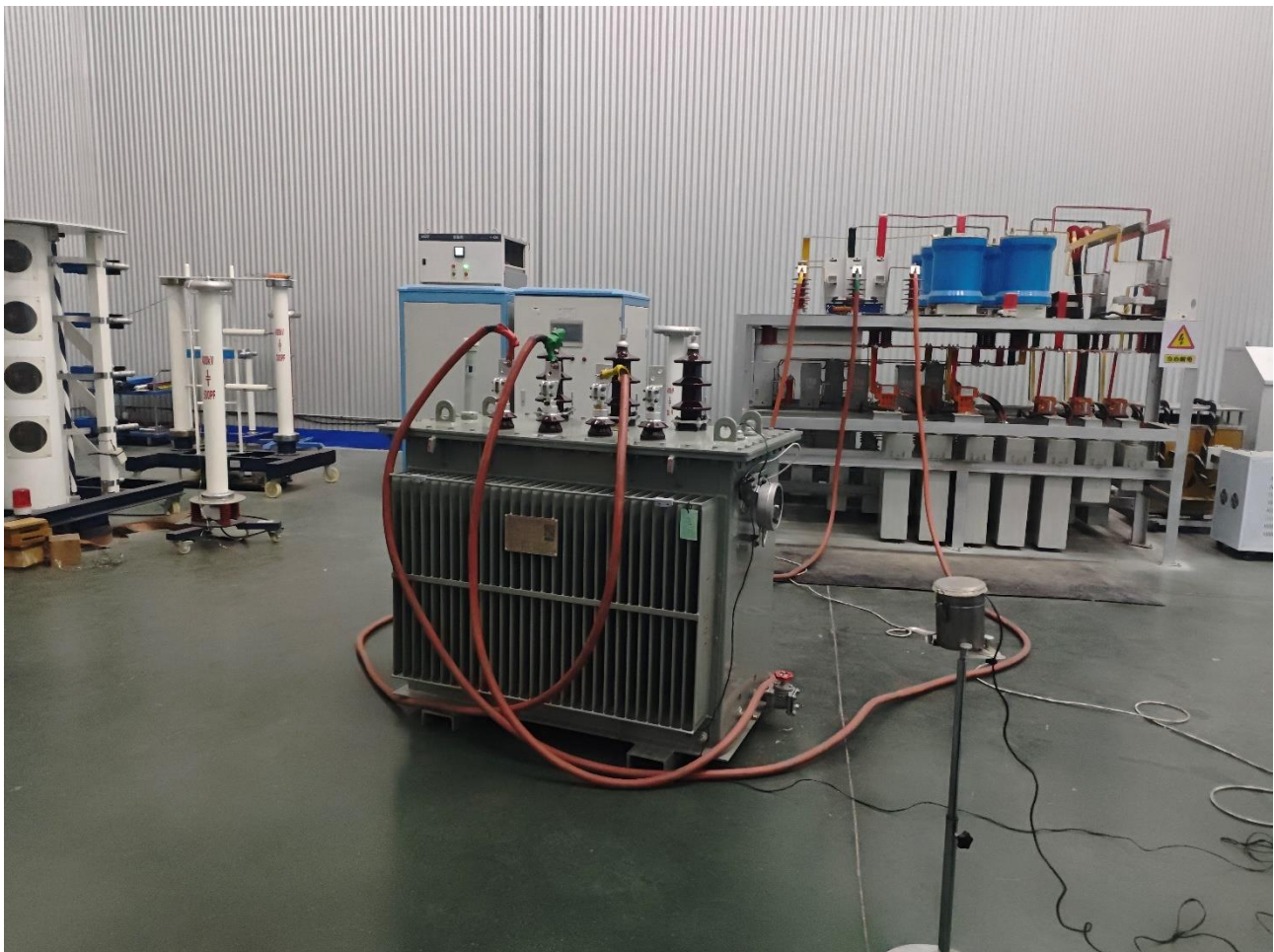
Test data

Measurement of no-load loss and current								
Tap position	Test voltage		Ratio of applied voltage to rated voltage	No-load current		No-load loss		
	Mean value (U')	r.m.s. value (U)		Mean value		Mean value (Pm)	Corrected value (P0)	Required value
	[V]	[%]		[A]	[%]	[W]	[W]	[W]
3	420.81	422.22	100	1.1939	0.14	511.71	510.00	≤540
	378.33	378.79	90	0.8571	0.10	402.69	402.20	--
	462.17	462.06	110	1.7867	0.21	643.82	643.97	--
Calculated equation: No-load current ratio = No-load current/Rated current of LV windings $P0 = Pm(1+d)$ $d = (U' - U)/U'$								

Result

Passed, the measured no-load loss and current at 100% of the rated voltage were within the specified tolerance. And the no-load and current at 90% and 110% of the rated voltage are only for information.

Test photograph



A.3 Determination of sound levels

Standard and date

Standard	EN 60076-10
Test date	2025-05-27

Environmental conditions

Ambient temperature	28.7 °C
Ambient air pressure	101.4 kPa
Relative humidity	27%

Characteristic test data

Serial number of power transformer	BY04101
Cooling method	ONAN
Rated power	630 kVA
Height of transformer oil tank	1.098 m

Requirement

Sound power level shall not exceed 45 dB(A) at ONAN condition.

Sound power calculation due to load current

In order to judge whether it is necessary to measure the ground sound level under load current, it can be roughly estimated by following equation:

$$L_{WA, Ir} \approx 39 + 18 \times \lg \frac{S_r}{S_p} \text{ for 50 Hz power frequency}$$

$$L_{WA, Ir} \approx 44 + 18 \times \lg \frac{S_r}{S_p} \text{ for 60 Hz power frequency}$$

Where,

$L_{WA, Ir}$ is the estimated A-weighted sound power level of the transformer at rated current and rated frequency at short-circuit condition.

S_r is the rated power in MVA.

S_p is the reference power (1 MVA).

Estimated sound level of this transformer (630 kVA): $L_{WA} = 39 + 18 \times \lg (0.63/1) = 35.4 \text{ dB(A)}$

The calculated values are more than 10 dB below the sound power level estimated at no-load excitation, therefore no need to test on load current.

Procedure

The acoustic sound level was measured at the selected locations, around the transformer, at half height at no-load condition.

The test was performed in the ONAN situation with rated voltage and the tap changer in position 3 and energized from the low voltage windings. The measuring points were measured at 0.3 meter from the principal radiating surface.

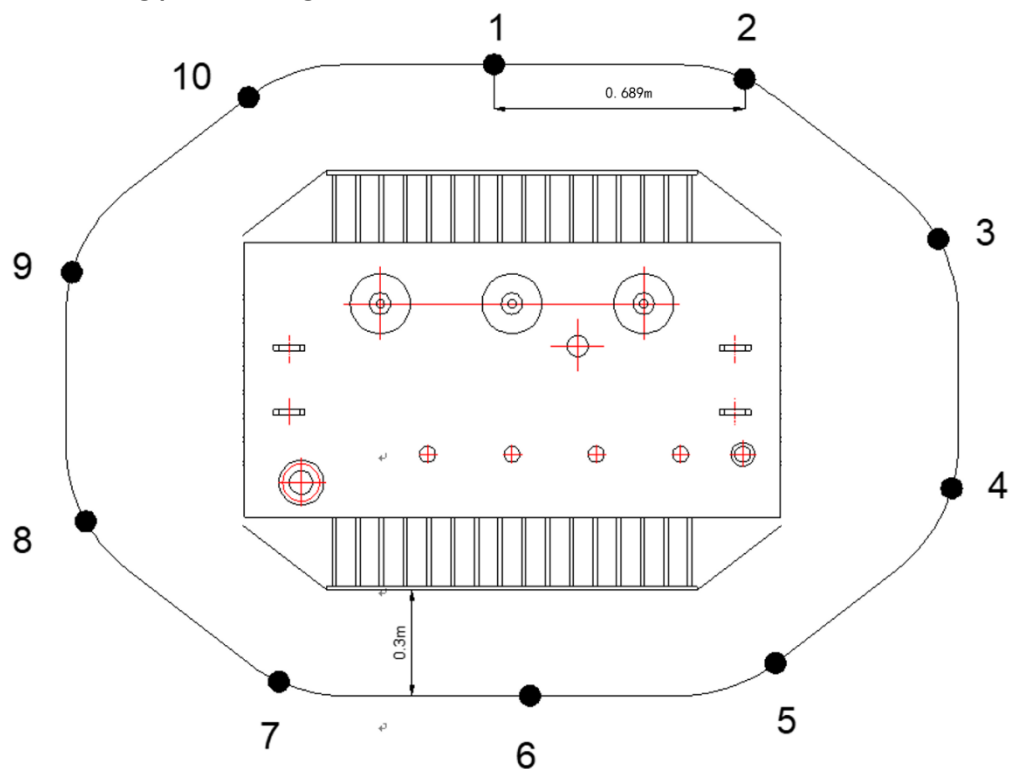
Test data

Determination of sound levels										
Height of transformer (h): <u>1098 mm</u>										
Perimeter of the outline (lm): <u>6.892 m</u>										
Measuring area proportion: $S = (h+0.3) \times lm = \underline{9.625 \text{ m}^2}$										
Environmental correction: $K = 10 \times \lg \left(1 + \frac{4}{A/S} \right) = \underline{0.39 \text{ dB}}$										
Average acoustic absorption coefficient: Hard floor: $\alpha = \underline{0.1}$; Wall and ceiling with acoustic damping material of up to 20 cm thickness: $\alpha = \underline{0.3}$										
Area of the partial surface of the test room: $S_{vi} = \underline{285 \text{ m}^2}$ for hard floor, 1277.8 m^2 for walls and ceilings										
Absorption Amount: $A = \alpha \times S_{vi} = \underline{411.84 \text{ m}^2}$										
Height of measuring point: <u>0.549 m</u>										
Average measuring point interval distance: <u>0.689 m</u>										
Test method: <u>Sound pressure method</u>										
☒ Sound power level L_{WA} (derived from sound pressure) ☐ Sound pressure level L_{PA} at measuring distance ☐ Sound pressure level L_{PA} at a specified distance										
Applied voltage [V]	Rated frequency [Hz]	Tap position	Distance [m]	In operation						
				No. Of fans	No. Of pumps					
420.01	50.0	3	0.3	N/A	N/A					
Sound pressure level measurement										
Measuring position	1	2	3	4	5	6	7	8	9	10
Before test of background noise	28.2	28.1	28.3	28.2	28.0	28.1	28.3	28.2	28.4	28.1
Sound pressure of object	35.3	35.1	35.2	35.5	35.2	35.4	35.2	35.0	35.3	35.5
After test of background noise	28.1	28.3	28.3	28.2	28.1	28.1	28.3	28.2	28.4	28.1
Average measured sound pressure level L_{PA0}				35.3 dB(A)						
Background sound pressure (before/after) L_{bgA}				28.3/28.2 dB(A)						
Environmental correction K				0.39 dB						
Corrected average sound pressure level L_{PA}				34.0 dB(A)						
Calculated sound power level L_{WA}				43.8 dB(A)						
Guaranteed sound power level L_{WA}				45 dB(A)						
Formula: $LPA = 10 \log(10^{0.1LPA0} - 10^{0.1LbgA}) - K$ $LWA = LPA + 10 \log(S/S0)$ S0 is equal to the reference area (1 m ²)										

Result

Sound power level was not exceed 45 dB(A) at ONAN condition.

Sound level measuring position diagram



Test photograph



Photo documentation

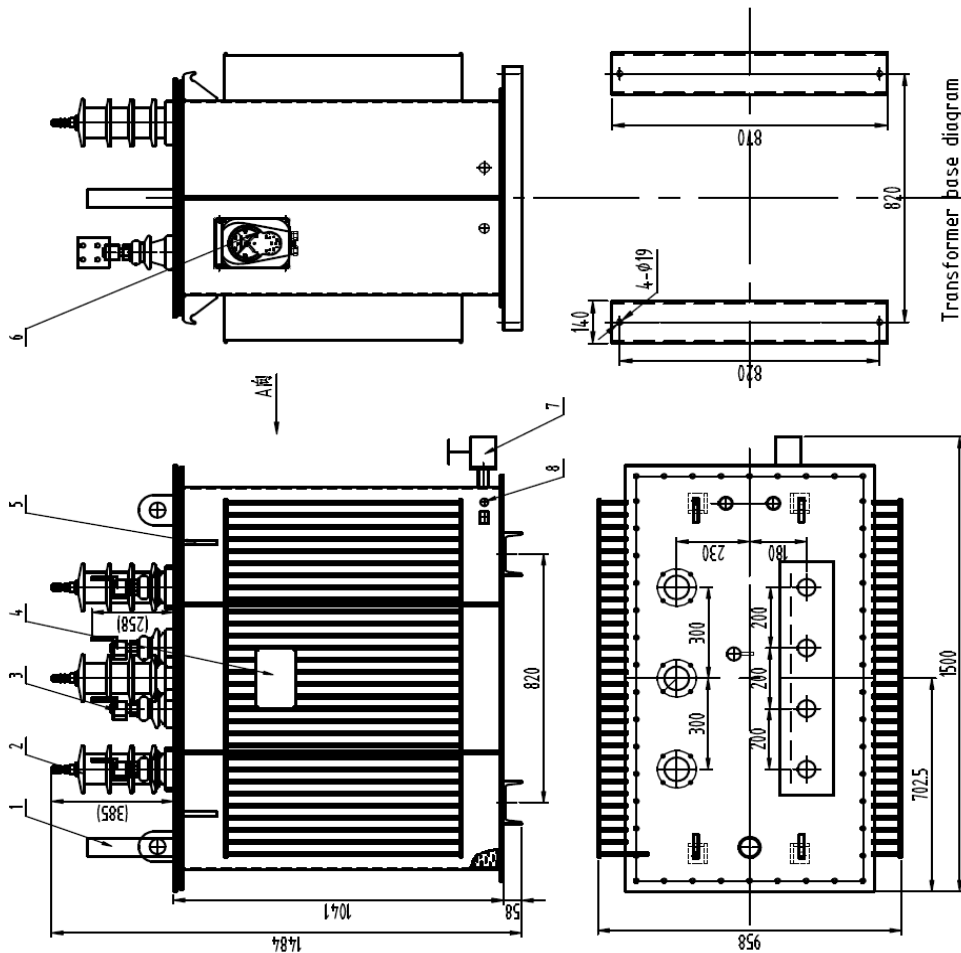




Drawings

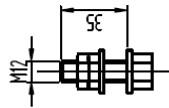
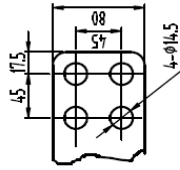
	OIL WEIGHT	BODY WEIGHT	ALL WEIGHT
(kg)	600	2150	3150

(kg)



LV bushing terminal block

HV bushing terminal block



Type: S-M-630/22-Tier2

Connection group : Dyn11

Impedance: 6.0%

Voltage regulation park: $22 \pm 2 \times 2.5\% / 0.42 \text{ kV}$

Color: RAL7033

8	Exhibit	1			
7	Valve	1			
6	Signaling Thermometer	1			
5	Lug	4			
4	Refinagrite	1			
3	L Vending	4			
2	Hydrating	3			
1	Technical oil-level indicator	1			
时	时	时	时	时	时

RB.710.8827.e